

Alexander Burns

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EDUCATION

Stanford University

M.S. AeroAstro

Palo Alto, CA

Expected June 2028

University of California, San Diego

B.S. Aerospace Engineering – Specializations: Astrodynamics, Controls | GPA: 3.96/4.00 | Magna Cum Laude

San Diego, CA

June 2026

Relevant Coursework: Aerodynamics, Aerospace Structures, Computational Fluid Dynamics, Controls, Fluid Mechanics, GNC, Heat Transfer, Material Science, Orbital Mechanics, Propulsion, Space Propulsion, Solid Mechanics, Thermodynamics, Vibrations

PROFESSIONAL EXPERIENCE

Starfish Space

Intern, GNC Engineer

Seattle, WA

June – September 2026

- Architected a generic attitude controller class and a composable PID controller in **Python/C++** for **Basilisk**, exposing control-law terms through YAML config to support entirely custom, Lyapunov, or other PID formulations without rebuilds or code changes.
- Supported power-aware proximity operations trajectory design by evaluating self and client solar array shadowing through the creation of a custom pyModel within **Basilisk**, and parallelized initial ray-tracing table computation for a 13x runtime speedup.
- Built a firmware-in-the-loop framework between Otter's thermal controller and **Basilisk's** 6-DOF Monte Carlo state propagation via a lockstep socket interface to validate thermal control capabilities across nominal and tumbling rendezvous.

Planet Labs

Intern, Spacecraft Systems Engineer

San Francisco, CA

June – September 2025

- Validated that previously un-characterized ferromagnetic materials on the Block II payload had insignificant impact on GNC control authority and satellite pointing by modeling theoretical fields in **FEMM** and computing an equivalent magnetic dipole moment.
- Developed a spacecraft power and drag simulation framework to evaluate generation and consumption profiles for LEO imaging satellites, accounting for sun fraction, attitude slews, operations, and GEO relay pointing constraints. Integrated results with Monte Carlo derived drag parameters from **OpenFOAM** to evaluate GEO pointing implications on lifetime prop consumption.

ViaSat

Intern, Satellite Operations Specialist

Carlsbad, CA

June 2023 – June 2025

- Performed trajectory optimization with the Flight Dynamics team to maintain orbital drift by predicting GEO perturbations and calculating station-keeping maneuvers to minimize prop consumption during nominal operations and transfer orbits.

ORGANIZATIONS

UCSD Large-Scale Design Optimization Laboratory | John Hwang

March 2025 – June 2026

- Applied **Nonlinear Model Predictive Control** to concurrently optimize quadrotor design and GNC algorithms, developing a holistic vehicle and trajectory model integrating system performance, mission requirements, and dynamic constraints.
- Improved solver convergence by implementing a segment-wise warm-starting strategy, enabling stable, computationally efficient solutions across randomized racecourse configurations that previously stalled or diverged.

UCSD Yonder Dynamics | URC Mechanical Team Member

October 2023 – April 2025

- Led design reviews using SolidWorks assemblies to ensure seamless integration, promoting cross-disciplinary collaboration.
- Designed, tested, and manufactured an improved rover elbow joint through Ansys Mechanical, achieving a 15% weight reduction by optimizing stress distributions and removing excess material, resulting in minimal backlash and improved load-bearing capacity.

PROJECTS

LEO Orbit Determination via Sequential State Estimation

March – June 2026

- Built Conventional and Extended Kalman Filters in MATLAB to estimate an 18-state LEO spacecraft trajectory, perturbation parameters, and ground station coordinates from range-rate tracking data, achieving sub-centimeter position uncertainty.
- Investigated EKF relinearization stability against the CKF's fixed-reference assumption, applying an observation warm-up to eliminate filter divergence and quantifying the effects of a priori covariance and station inclusion on estimation accuracy.

Gradient-Free Optimization of a Blended Wing Body

January – March 2026

- Implemented **Particle Swarm Optimization** with global-best topology to maximize cruise L/D of a BWB aircraft subject to structural, stability, and airport wingspan constraints – achieving a constraint-satisfying L/D of 25.30 via bound refinement.
- Characterized design space sensitivity through a latin hypercube **Monte Carlo** simulation across $\pm 40\%$ design parameter variations, revealing that overstressed configurations emerge before wingspan constraints, eliminating the need for a folding wing design.

SKILLS

Software: Basilisk, Python, C++, MATLAB SolidWorks, Ansys Fluent (Certified)/Mechanical, JavaScript, Linux, Office, JIRA, Git